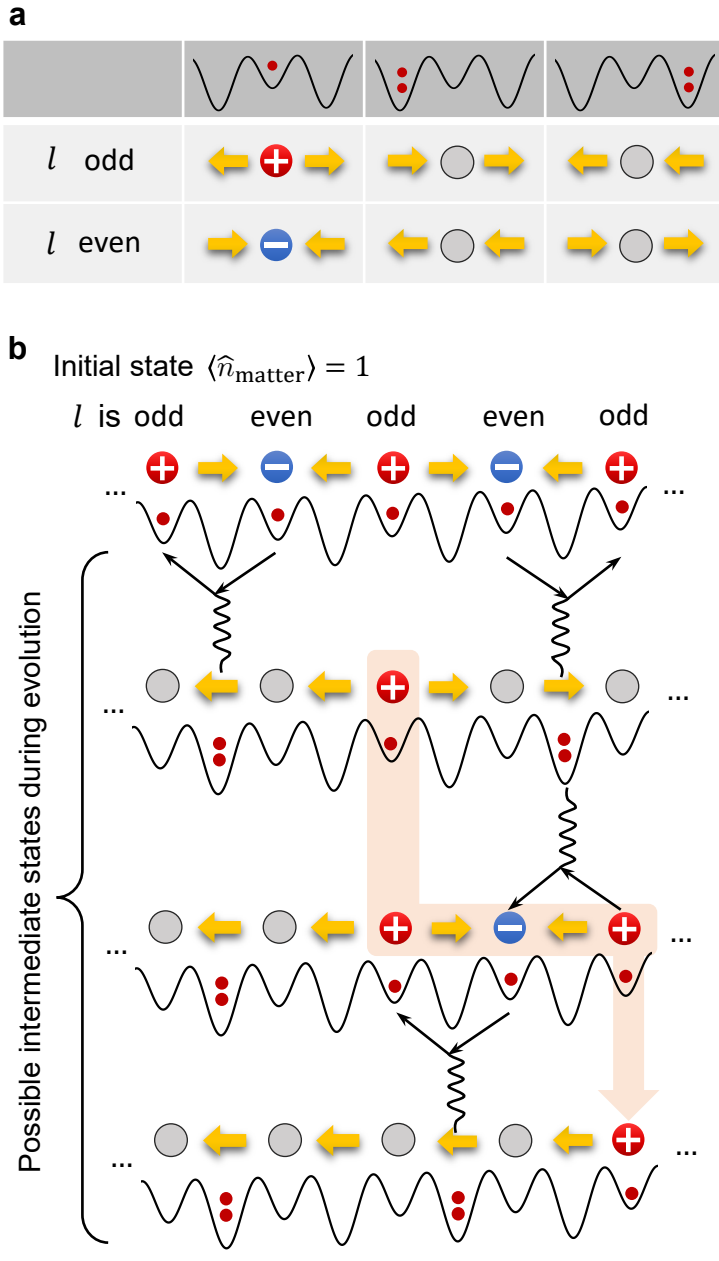

a
l odd
l even
b
Initial state $\langle \hat{n}_{\mathrm{matter}} \rangle = 1$
l is odd
even
odd
even
odd
Possible intermediate states during evolution